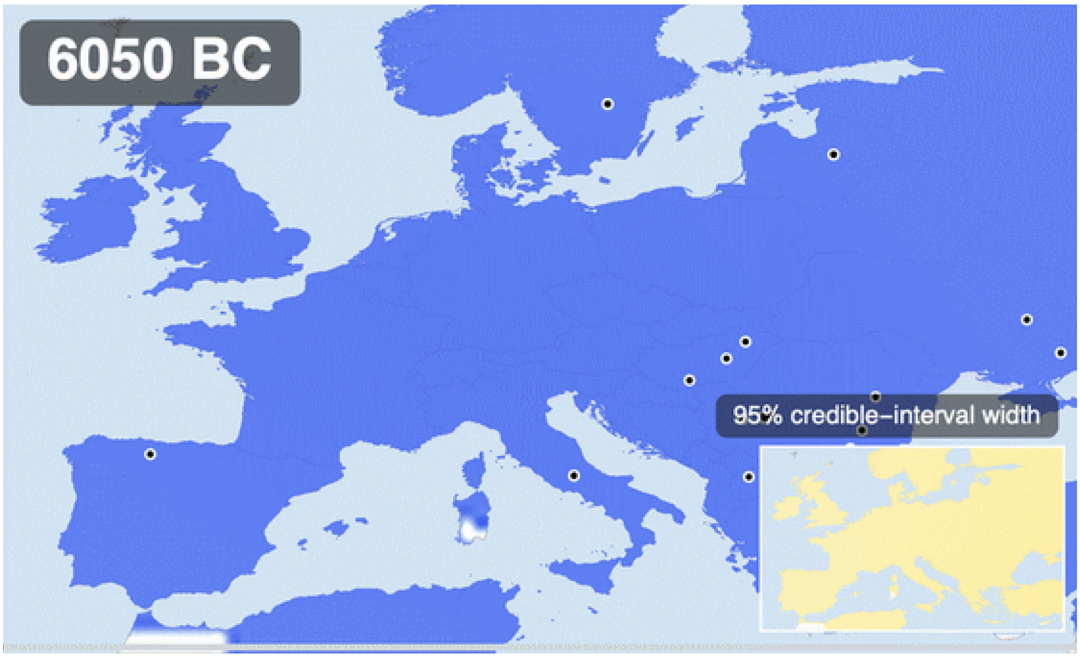
Watching evolution happen: lactase persistence in 8,000 years of ancient DNA

From raw ancient genotypes to a spatial diffusion–selection model, fitted with sequential Monte Carlo ABC — with the uncertainty carried all the way into the pictures



Posterior mean frequency of the lactase–persistence allele (rs4988235–A) across Europe, 8000 BC to today, on the TemperatureMap scale. White–ringed dots are the ancient individuals dated within ±400 years of the moving window; the inset tracks the 95% credible–interval width — where the map is honest about not knowing. Generated by ExportHeroAnimation in the companion package.

Around nine thousand years ago, everyone in Europe — every single adult — lost the ability to digest milk sugar a few years after weaning, just as almost all mammals do. Today, most northern Europeans can drink a glass of milk without a second thought. The switch that changed, a single letter of DNA about 14,000 base pairs upstream of the lactase gene, is the strongest signal of recent natural selection anywhere in the human genome — and thanks to ancient DNA we no longer have to infer its history from modern variation: we can simply watch it happen, skeleton by dated skeleton, across eight millennia.

Two recent results make this more interesting than the textbook just-so story. **Evershed et al. (2022, Nature)** assembled the largest set of prehistoric milk-fat residues from pottery and showed that Europeans were dairying enthusiastically for thousands of years while the allele stayed rare — and that a selection model tracking milk use explains the allele's trajectory *no better* than plain uniform selection since the Neolithic. **Burger et al. (2020, Current Biology)** showed the flip side: from a Bronze Age battlefield frequency of about 7%, reaching today's values requires a selection coefficient of roughly 0.06 per generation over the last 3,000 years — enormous by the standards of human evolution. Milk was everywhere, the allele was nowhere, and then, quite suddenly in evolutionary terms, it was everywhere too. Why?

This post does not settle that debate. It does something more modest and, I hope, more reusable: it builds the *entire* analysis pipeline in Wolfram Language, from the public ancient-genotype workbook to a calibrated spatial model — and treats uncertainty as a first-class output at every step. We will parse and clean 2,999 ancient genotype records, reproduce the classic regional frequency curves with proper binomial likelihoods, build a spatial diffusion–selection model on a landmasked grid of Europe, fit it with sequential Monte Carlo approximate Bayesian computation (10,000 forward simulations), check it with posterior predictive simulation and two kinds of held-out validation, stress-test the priors, and render the posterior — mean *and* credible-interval width — as the animation above. Every figure sits directly below the code that makes it, and the whole thing reruns from two wolframscript commands (§13).

All the heavy lifting lives in the companion package `LactasePersistenceSpatial.wl` — the only file you need beside this notebook (plus an internet connection for the first data download). Put the two files in one folder and evaluate: the setup cell finds the package next to the notebook (or in `src/` when you are inside the cloned repository), the data download and every output recreate themselves relative to that folder, and the expensive SMC posterior reloads from stored result tables when present and refits from scratch when not. Every output you see below was produced by running this notebook top to bottom with wolframscript.

1. Setup and data

The raw data are the public GLAD ancient-genotype workbook (Global Lactase persistence Association Database), derived from the Allen Ancient DNA Resource v44.3 and used for the Evershed et al. (2022) analysis. `RetrieveRawData` downloads it once, records a SHA-256 checksum and a provenance manifest, and marks the file read-only; `WriteProcessedData` parses, cleans, and writes tidy CSVs with provenance attached.

```
In[∘]:= nbdir = NotebookDirectory[];
root = If[FileExistsQ[FileNameJoin[{ParentDirectory[nbdir], "src", "LactasePersistenceSpatial.wl"}]],
  ParentDirectory[nbdir], nbdir];
SetDirectory[root];
Get[First[Select[
  {FileNameJoin[{root, "src", "LactasePersistenceSpatial.wl"}],
   FileNameJoin[{nbdir, "LactasePersistenceSpatial.wl"}]}, FileExistsQ]]];
```

```
In[∘]:= raw = RetrieveRawData[Directory[]];
processed = WriteProcessedData[Directory[], raw];
samples = processed["CalledSamples"];
Length[samples]
```

```
Out[∘]=
1785
```

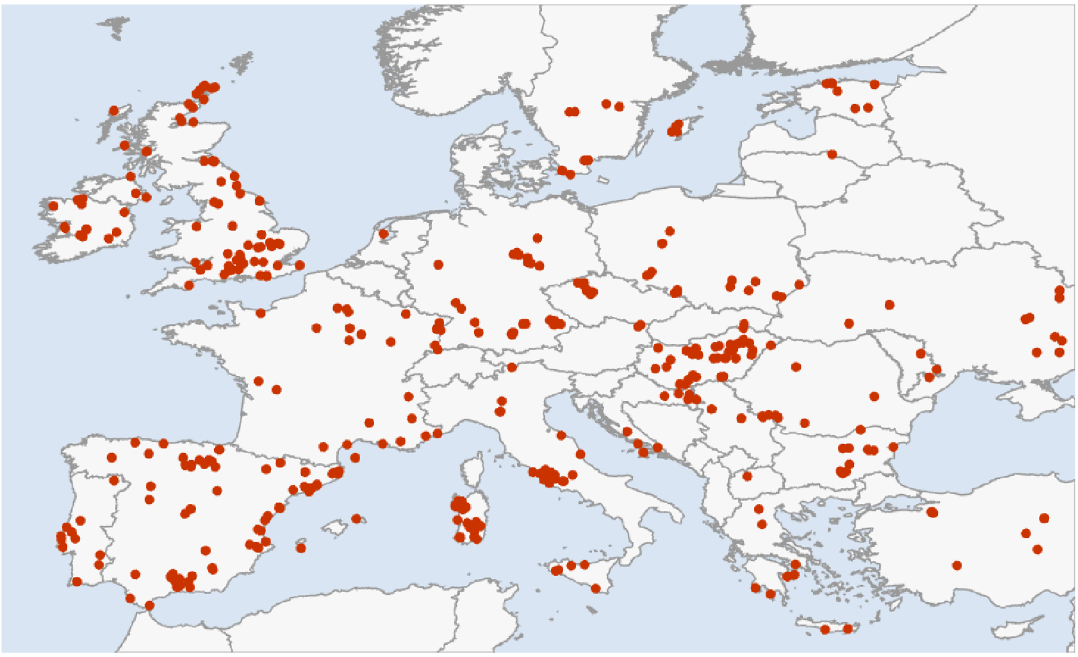
Parsing ancient genotypes is where silent errors live, so the cleaning rules are explicit and tested: genotype strings are normalized so that A — and the strand-flipped T, which occurs in exactly 3 heterozygous calls — count as the derived (persistence) allele and G/C as ancestral; a single-letter pseudo-haploid call contributes one called allele and a diploid call two; coordinates accidentally recorded in millidegrees (37724 for Sicily) are rescaled; and every sample is assigned to one of four coarse analysis regions echoing the published framing. Of 2,999 rows, 1785 have a genotype call plus usable age and location.

In[∘]:=
```
Counts[#["Region"] & /@ samples]
```

Out[∘]=
```
<| Rhine–Danube → 341, Outside Europe → 681,
  Other Europe → 50, Mediterranean → 486, British Isles → 171, Baltic → 56 |>
```

In[∘]:=
```
GeoListPlot[GeoPosition[{#["Latitude"], #["Longitude"]} & /@ samples],
 GeoRange -> {{34, 63}, {-12, 36}}, GeoProjection -> "Equirectangular",
 PlotStyle -> Directive[Opacity[0.45], RGBColor[0.153, 0.51, 0.64], PointSize[0.005]],
 GeoBackground -> GeoStyling[{"CountryBorders", "Land" -> GrayLevel[0.97], "Ocean" -> RGBColor[0.8
 ImageSize -> 620]
```

Out[∘]=

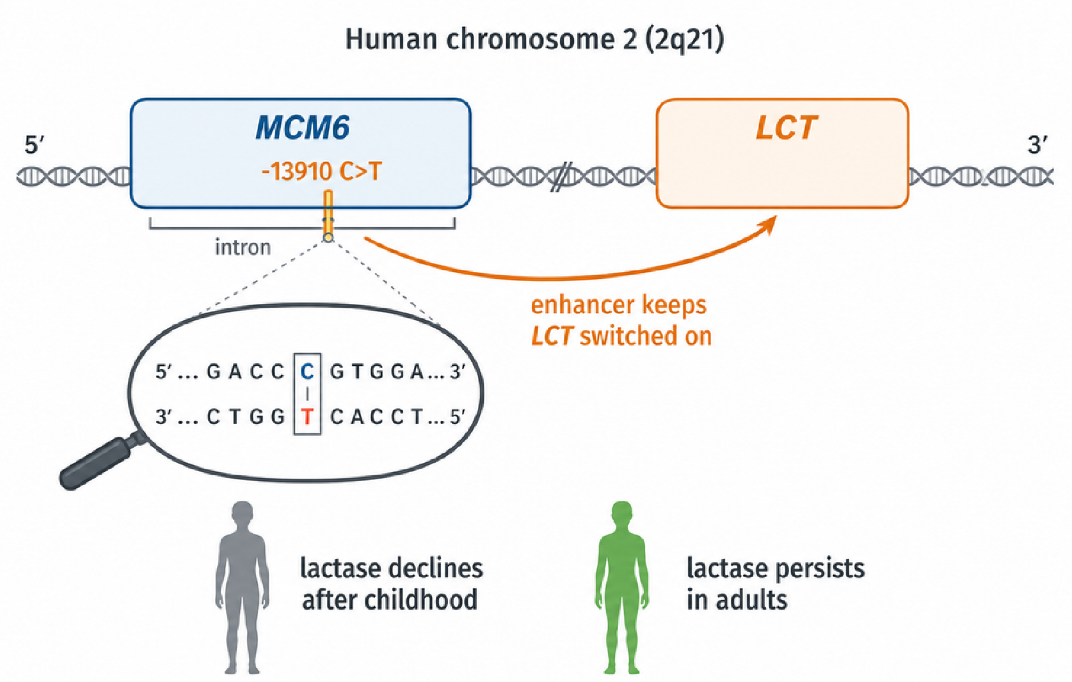


Where the 1785 usable ancient individuals come from. Density is wildly uneven in space and time — the Mediterranean is rich, the Baltic sparse, and almost nothing is younger than 2,500 years — and that unevenness is exactly why the model must carry its uncertainty around explicitly.

2. What is lactase persistence?

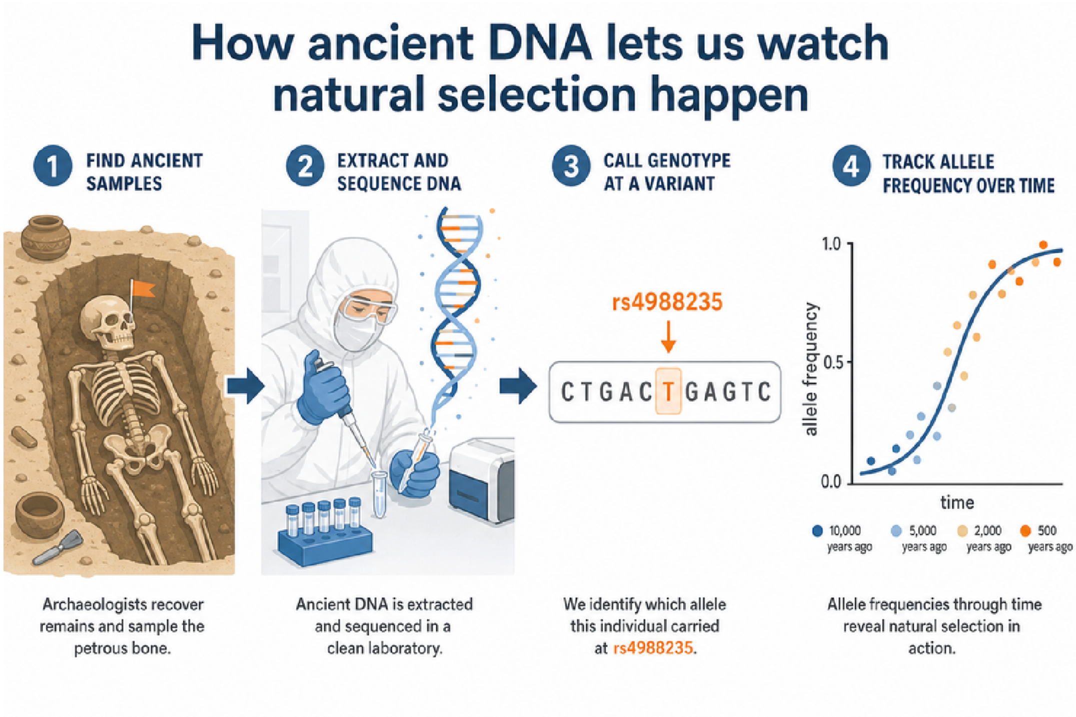
Lactose, the sugar in milk, cannot be absorbed directly: it must first be split into glucose and galactose by the enzyme *lactase*, produced in the lining of the small intestine and encoded by the `LCT` gene. In the ancestral mammalian program, lactase production is switched off after weaning — an adult wolf, cow, or (ancestrally) human who drinks milk gets bloating, cramps, and worse, because undigested lactose ferments in the colon. The European persistence variant is not a change to lactase itself: it is a single C→T substitution 13,910 bases upstream, inside an intron of the neighbouring gene `MCM6`, that acts as an enhancer and simply keeps the LCT switch turned on into adulthood. On the reported strand of the association database this is the G→A change at `rs4988235` — one letter, one glass of milk, and (as we will see) several percent of extra surviving children per generation at its peak.

> ***External asset.*** The gene schematic was generated outside Wolfram Language with OpenAI's `gpt-image-2` model; the prompts and the one-off Python driver are checked into `docs/images/figures-generated/` of the project repository. It is a static illustration; there is no WL code to run for this figure.



A crucial subtlety for what follows: the variant is effectively *dominant* — one copy is enough to digest milk — and it rides on a long shared haplotype, which is how it was first detected from modern data alone. But modern data compress history: they can say the allele rose fast, not *when*. Ancient DNA turns the same question into direct observation.

> ***External asset.*** The ancient-DNA workflow illustration was generated outside Wolfram Language with OpenAI's `gpt-image-2` model; the prompts and the one-off Python driver are checked into `docs/images/figures-generated/` of the project repository. It is a static illustration; there is no WL code to run for this figure.



3. The regional story: four binomial logistic fits

Before any spatial machinery, the honest first step is to reproduce the standard non-spatial description: for each region, a logistic trajectory `logit p(t) = α + β (10000 − t)/1000` fitted by maximizing the exact binomial log-likelihood over every called allele — no time-binning enters the fit; the bins you see in the figure are display only, each carrying a 95% Wilson score interval and sized by its allele count. Standard errors come from the numerical Hessian at the optimum, and the fitter flags any solution that lands on a parameter bound. That flag is not decoration: an early version of this project silently reported a boundary solution for the British Isles, and the only symptom was a suspiciously round α. Fits you cannot audit are fits you cannot trust.

In[∘]:=
```
fits = FitAllRegionalLogistics[samples];
Grid[
 Prepend[
  Table[With[{f = SelectFirst[fits, #["Region"] === r &]},
    {r, f["SampleCount"], f["CalledAlleles"], f["DerivedAlleles"],
     Row[{NumberForm[f["Alpha"], {5, 2}], " ± ", NumberForm[f["AlphaSE"], {4, 2}]}],
     Row[{NumberForm[f["BetaPerKyrTowardPresent"], {4, 2}], " ± ", NumberForm[f["BetaSE"], {4, 2}]}],
     NumberForm[f["SelectionPerGenerationApprox"], {4, 3}]}],
   {r, MajorRegions[]}],
  Style[#, Bold] & /@ {"region", "samples", "alleles", "derived", "α", "β per kyr", "≈s per gen"}],
 Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.2, 0.6},
 Background -> {None, {GrayLevel[0.93], None}}]
```

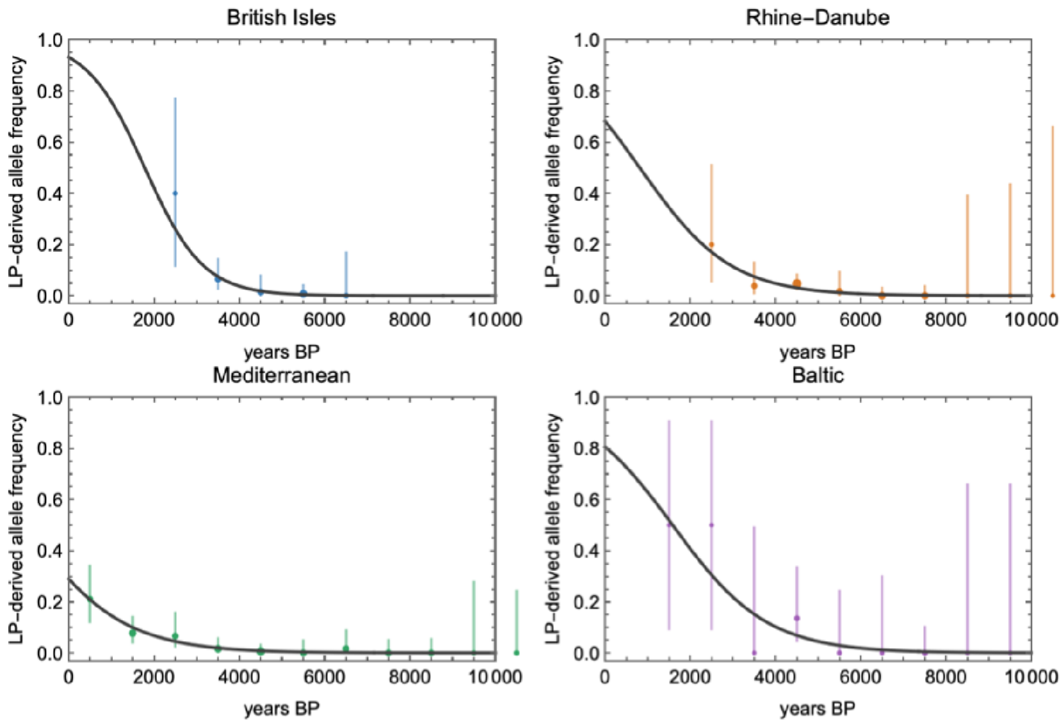
Out[∘]=

region	samples	alleles	derived	α	β per kyr	≈s per gen
British Isles	171	304	9	−12.02 ± 3.24	1.46 ± 0.51	0.041
Rhine–Danube	341	591	16	−8.60 ± 1.58	0.94 ± 0.27	0.026
Mediterranean	486	837	28	−9.54 ± 1.35	0.86 ± 0.16	0.024
Baltic	56	90	5	−7.55 ± 2.01	0.90 ± 0.31	0.025

With a 28-year generation time, a slope β on the logit scale translates to an approximate per-generation (genic, codominant) selection coefficient s ≈ 28β/1000. The four regions land between 0.024 and 0.041 — the familiar few-percent picture, strongest in the north-west, and comfortably bracketing Burger et al.'s 0.06 once you note their estimate concentrates on the last three millennia while these slopes average over eight.

In[∘]:=
```
regionalOutputs = ExportRegionalFitOutputs[Directory[], samples, fits];
Import[regionalOutputs["RegionalFigure"]]
```

Out[∘]=

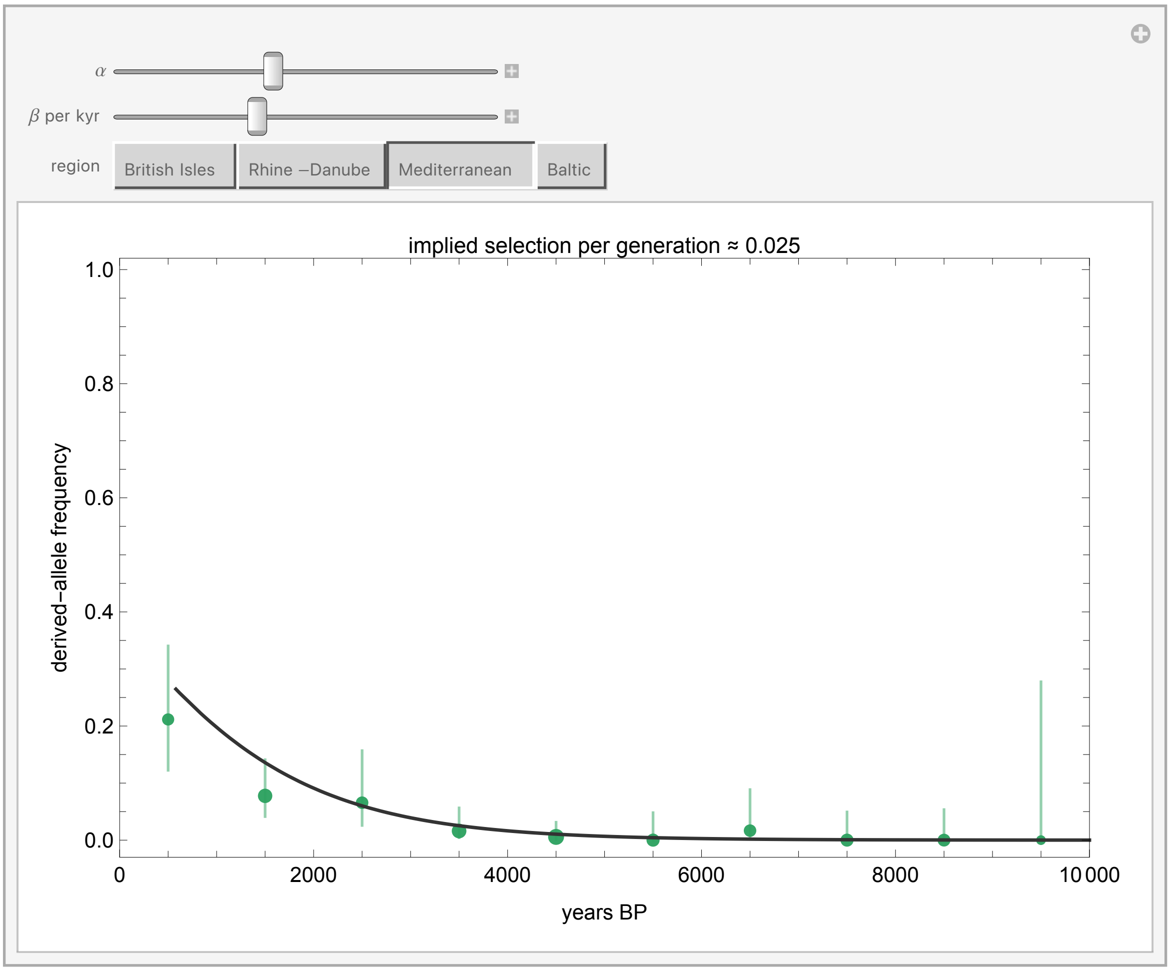


Observed binned frequencies (Wilson 95% intervals, point area ∝ allele count) with the fitted logistic curves. Note the Mediterranean's leisurely rise against the British and Baltic sprints — and note also how far right of the last data point every extrapolation to the present runs. Everything after ~2,000 BP is extrapolation, and the spatial model below inherits that caveat.

Rather than take my word for how sensitive these curves are, drag them yourself. The Manipulate below overlays an adjustable logistic on each region's binned data — the label converts your β into an implied selection coefficient as you drag. It takes about ten seconds of play to internalise two things: how narrow the corridor of plausible β is for the data-rich Mediterranean, and how wide it is for the Baltic.

In[∘]:=
```
LogisticExplorer[samples]
```

Out[∘]=



4. A spatial diffusion–selection model

Regions are administrative conveniences; alleles do not respect them. The spatial model discretizes Europe (35–63°N, 12°W–35°E) into a 2° grid, keeps only cells whose centres are on land (191 of them — so Britain stays rook-connected to the continent across the Dover cells, while Ireland becomes an island in the model's adjacency too, a caveat worth knowing), and lets the local allele frequency evolve in 250-year steps from 10,000 BP:

```
p[i, t+dt] = p[i, t] + g ( (s0 + sDairy D[i, t] m[region i]) p (1 - p)
                           + mig mean[ p[j, t] - p[i, t], j adjacent to i ] )

g = dt / 28 generations per step
```

Selection has a baseline component s0 and a component sDairy switched on by a smoothly varying dairying-onset field D(i, t) — built by inverse-distance interpolation from six regional anchor dates (Mediterranean ~8200 BP through Baltic ~5600 BP) rather than four flat plateaus, so the covariate has genuine spatial texture. Per-region multipliers m let the north run hotter than the south, and the migration term exchanges frequency with adjacent land cells. Here is the model's actual geography — the land cells it evolves on, coloured by when dairying arrives in each:

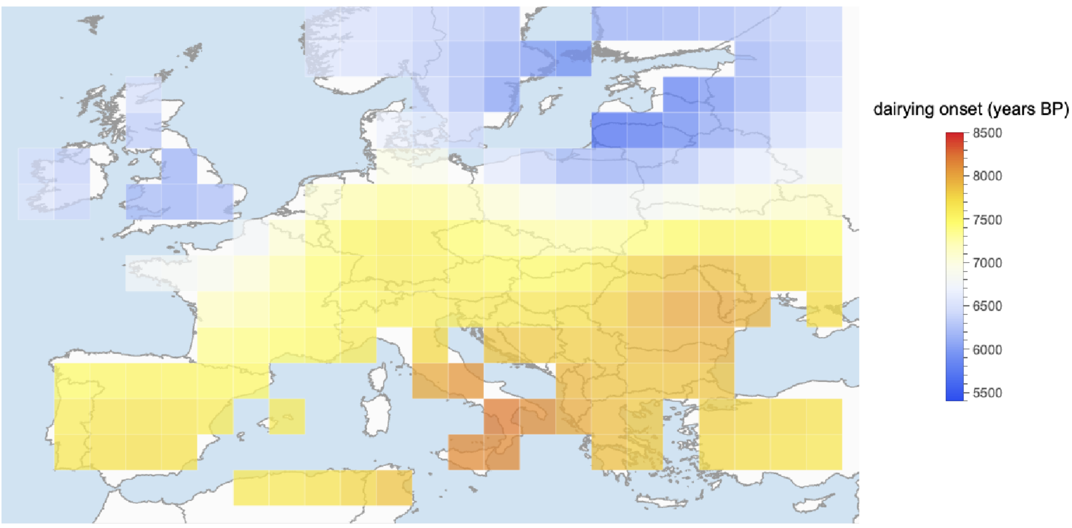
In[•]:=
```
grid = BuildEuropeGrid[];
Length[grid]
```

Out[•]=

191

In[•]:=
```
Legended[
 GeoGraphics[
  Flatten@Table[
   {GeoStyling[Opacity[0.75,
     ColorData["TemperatureMap"][Rescale[c["DairyingOnsetBP"], {5400, 8500}]]]],
    EdgeForm[Directive[White, Opacity[0.6], AbsoluteThickness[0.3]]],
    GeoPolygon[{{c["Latitude"] - 1, c["Longitude"] - 1}, {c["Latitude"] - 1, c["Longitude"] + 1},
      {c["Latitude"] + 1, c["Longitude"] + 1}, {c["Latitude"] + 1, c["Longitude"] - 1}}]},
   {c, grid}],
  GeoRange -> {{34, 63}, {-12, 36}}, GeoProjection -> "Equirectangular",
  GeoBackground -> GeoStyling[{"CountryBorders",
   "Land" -> GrayLevel[0.985], "Ocean" -> RGBColor[0.82, 0.89, 0.95]}],
  ImageSize -> 640],
 BarLegend[{(ColorData["TemperatureMap"][Rescale[#, {5400, 8500}]] &), {5400, 8500}},
  LegendLabel -> "dairying onset (years BP)"]]
```

Out[•]=

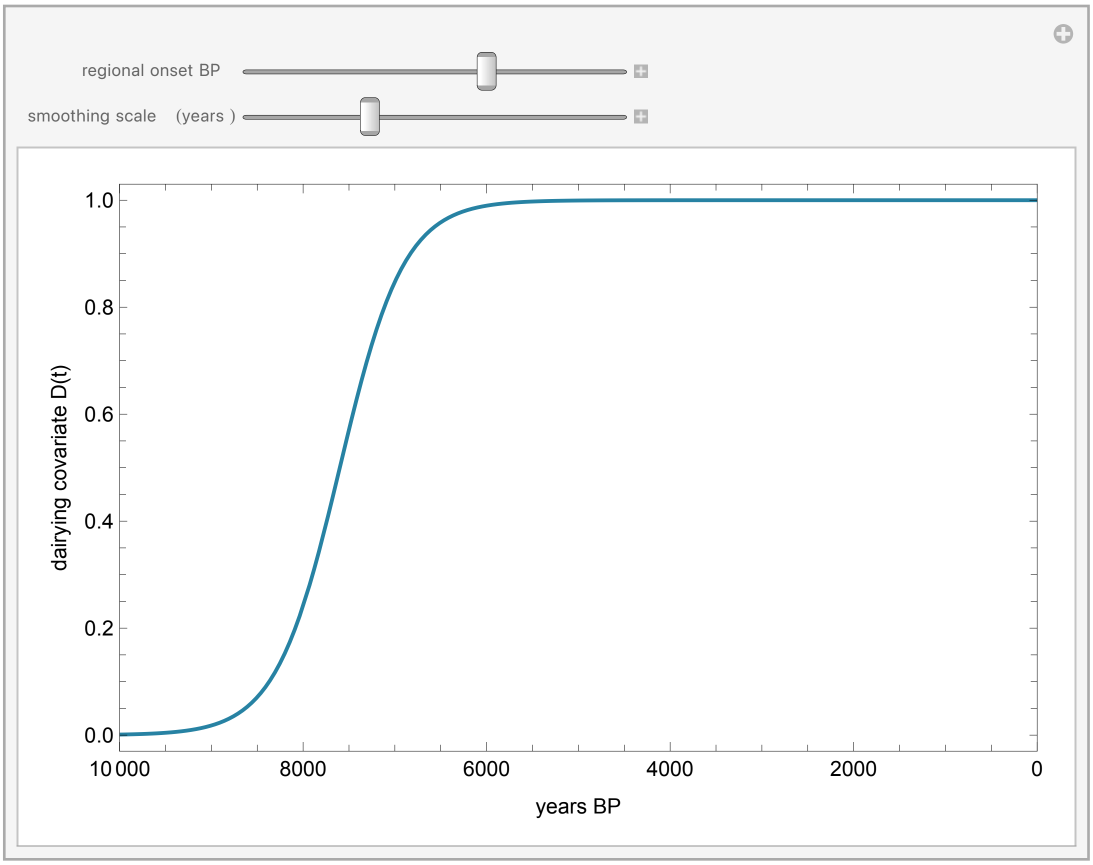


The model's spatial skeleton: the 2° land cells (sea cells are removed by an elevation test, so diffusion respects coastlines — Britain stays connected across the Dover cells, Ireland is an island in the adjacency too), coloured by the smooth dairying-onset field. Red is early dairying (Mediterranean and the southeast, ~8,200 BP), blue is late (the Baltic, ~5,600 BP). This field is what the sDairy component of selection switches on, cell by cell, as the simulation runs.

The covariate's shape in time is worth ten seconds of interaction too:

In[∘]:=
```
DairyingCovariateExplorer[]
```

Out[∘]=



One forward simulation of this model costs about 70 ms:

In[∘]:=
```
AbsoluteTiming[
  SimulateSpatialTrajectory[
   <|"InitialFrequency" -> 0.003, "SelectionBase" -> 0.002,
    "SelectionDairying" -> 0.02, "Migration" -> 0.003|>, grid];
 ][[1]]
```

Out[∘]=

0.052564

— and that number is the entire strategy. When a simulator is this cheap, you do not need to linearize, approximate, or hand-wave the likelihood: you can afford tens of thousands of honest forward runs and let simulation-based inference do the rest. The model deliberately keeps demography implicit; it is a reaction–diffusion caricature whose job is to let the *data* say how much spatial structure, movement, and dairying-modulation they can constrain — not to smuggle conclusions in through the architecture.

External asset. The Neolithic dairying scene was generated outside Wolfram Language with OpenAI's `gpt-image-2` model; the prompts and the one-off Python driver are checked into `docs/images/figures-generated/` of the project repository. It is a static illustration; there is no WL code to run for this figure.

5. Inference: sequential Monte Carlo ABC

Approximate Bayesian computation asks the only question a simulator can answer: which parameters produce synthetic data that look like the real data? "Look like" is made precise through summary statistics; here, two families. First, the called-allele-weighted regional time-binned frequencies (39 bins). Second — because a spatial model scored only on regional aggregates can never learn about migration — each sample is matched to its nearest grid cell and time step, and pooled north–south and west–east frequency contrasts over the last 4,000 years are computed *identically* for observed alleles and model expectations. The distance is a weighted root-mean-square over all of it.

Plain rejection ABC wastes almost every simulation once the tolerance gets interesting, so the fit uses sequential Monte Carlo: a population of 400 particles is filtered through 5 generations of shrinking tolerance (each generation's ϵ is the median of the previous population's distances), with Gaussian perturbation kernels and importance weights. One implementation lesson worth passing on: with a 10-dimensional box prior, perturbing in the raw coordinates throws ~97% of proposals out of the box (0.7 to the 10th ≈ 0.03). The sampler therefore works in logit-transformed coordinates, where a uniform prior becomes an exact product of standard logistic densities, every proposal is automatically valid, and the importance weights stay closed-form.

In[ ]:=
```
smc = LoadOrRunSMCABC[Directory[], samples, grid];
Grid[
 Prepend[
  Table[{k, NumberForm[smc["EpsilonHistory"][[k]], {5, 4}],
    NumberForm[100. smc["AcceptanceHistory"][[k]], {4, 1}],
    NumberForm[smc["ESSHistory"][[k]], {5, 1}]}, {k, Length[smc["EpsilonHistory"]]}],
  Style[#, Bold] & /@ {"generation", "ϵ", "acceptance %", "ESS"}],
 Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.5, 0.6},
 Background -> {None, {GrayLevel[0.93], None}}]
```

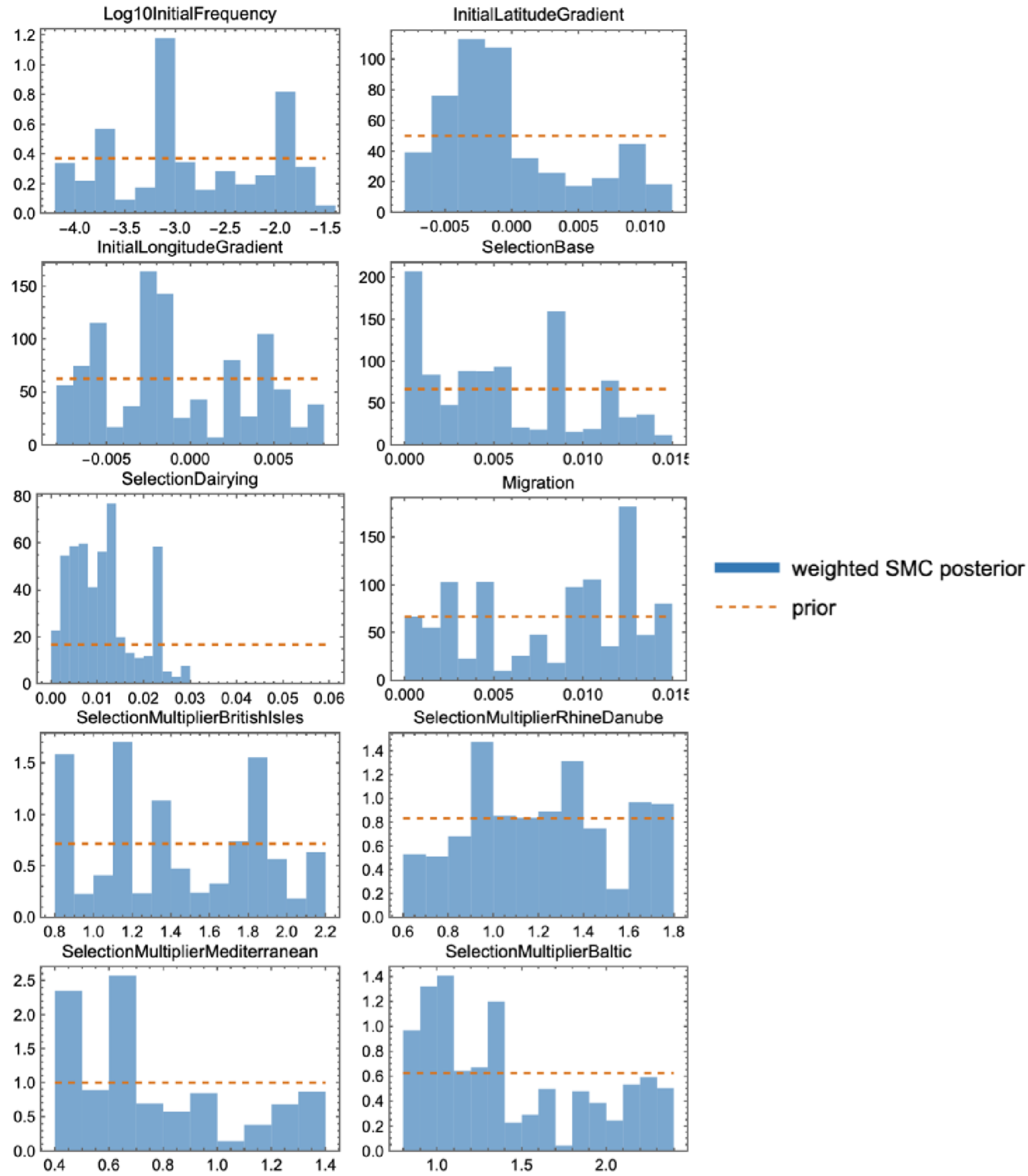
Out[ ]=

generation	ϵ	acceptance %	ESS
1	0.2582	100.0	400.0
2	0.0530	25.0	15.7
3	0.0483	25.0	46.0
4	0.0439	25.0	16.0
5	0.0408	12.5	26.6

The tolerance falls from 0.258 (the prior median — i.e. no information) to 0.041 over 10,000 simulations. The final effective sample size, 26.6 of 400, deserves a sentence rather than a footnote: importance-weight degeneracy is the known cost of this weight formula whenever the posterior is much tighter than the prior, every interval in this post is a weighted quantile and correspondingly approximate, and the honest cure (an SMC sampler with MCMC rejuvenation moves) is on the to-do list in §11. Reporting ESS is what separates a posterior from a scatter plot.

In[ ]:=
```
draws = ResamplePosterior[smc, 100];
smcOutputs = ExportSMCOutputs[Directory[], samples, grid, smc, draws];
Import[smcOutputs["ParameterFigure"]]
```

Out[ ]=



Weighted SMC posterior (blue) against the flat prior (dashed orange) for all ten parameters. Prior overlays are the fastest honesty check in Bayesian workflow: one glance separates the dimensions the data actually moved (initial frequency, dairying-modulated selection, the northern multipliers) from the ones still wearing their priors (the initial spatial gradients).

6. What the posterior says

In[•]:=
```
quantiles = PosteriorParameterQuantiles[smc];
Grid[
 Prepend[
  Table[With[{q = SelectFirst[quantiles, #["Parameter"] === p &]},
    {p, NumberForm[q["Median"], {6, 4}],
     Row[{"[", NumberForm[q["Lower95"], {6, 4}], ", ", NumberForm[q["Upper95"], {6, 4}], "]"}]}],
   {p, {"Log10InitialFrequency", "SelectionBase", "SelectionDairying", "Migration",
     "SelectionMultiplierBritishIsles", "SelectionMultiplierRhineDanube",
     "SelectionMultiplierMediterranean", "SelectionMultiplierBaltic"}}],
  Style[#, Bold] & /@ {"parameter", "median", "95% interval"}],
 Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.5, 0.6},
 Background -> {None, {GrayLevel[0.93], None}}]
```

Out[•]=

parameter	median	95% interval
Log10InitialFrequency	−3.0907	[−4.1812, −1.6381]
SelectionBase	0.0044	[0.0001, 0.0138]
SelectionDairying	0.0103	[0.0016, 0.0258]
Migration	0.0090	[0.0002, 0.0148]
SelectionMultiplierBritishIsles	1.3773	[0.8115, 2.1895]
SelectionMultiplierRhineDanube	1.2170	[0.6320, 1.7686]
SelectionMultiplierMediterranean	0.6253	[0.4093, 1.3759]
SelectionMultiplierBaltic	1.2748	[0.8123, 2.3342]

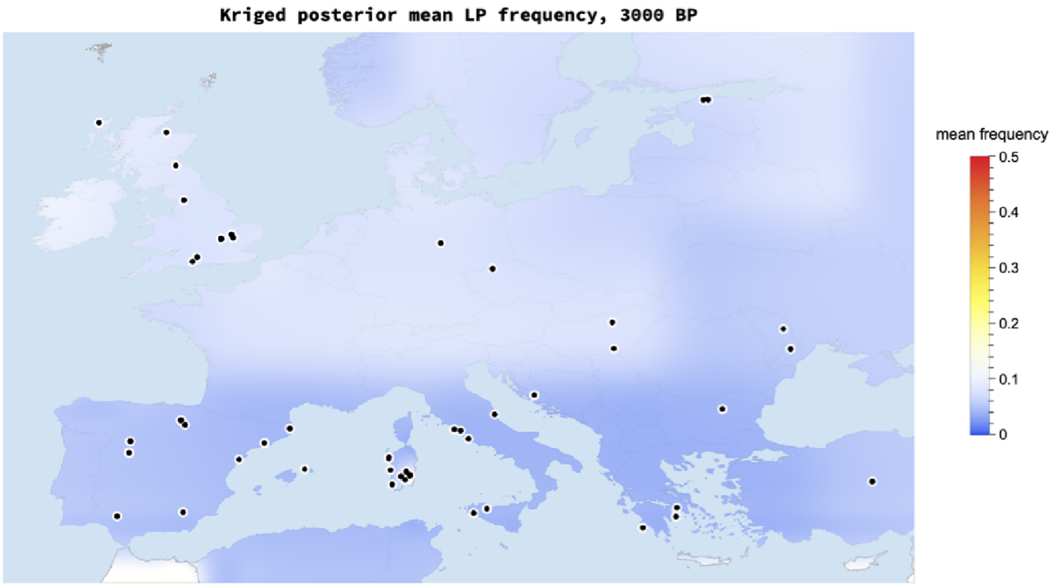
Three readings, in decreasing order of confidence. **First**, the total effective selection in a northern dairying cell at the posterior median — s0 + sDairy × multiplier ≈ 0.019 per generation — sits squarely in the published few-percent range, with the northern multipliers pulled above one exactly where the logistic slopes were steepest. **Second**, the posterior does *not* cleanly separate dairying-modulated from baseline selection. This is not a failure of the sampler; it is the geometry of the problem. The dairying covariate saturates to one within a few centuries of onset, so for most of the simulated period sDairy acts almost exactly like uniform selection; only the early Neolithic carries contrast between the two. That the ancient genotypes accommodate both a mostly-uniform and a mostly-dairy-modulated reading is this little model's echo of Evershed et al.'s conclusion at Nature scale: milk exploitation is not required to explain the trajectory. **Third**, migration is only weakly identified — §9 quantifies exactly how weakly, instead of letting a prior masquerade as a discovery.

7. The posterior on the map — and its uncertainty

Every map below is a posterior functional: 100 equally-weighted posterior draws, each pushed through the forward model, summarized per cell, then interpolated for display by ordinary kriging (range 3.5°, nugget 0.02) and cut to the coastline. Kriging here is a display courtesy, not an inferential claim — the 2° cells remain the units that saw data. Colors are the TemperatureMap scale with a square-root stretch so the long rare-allele centuries stay legible.

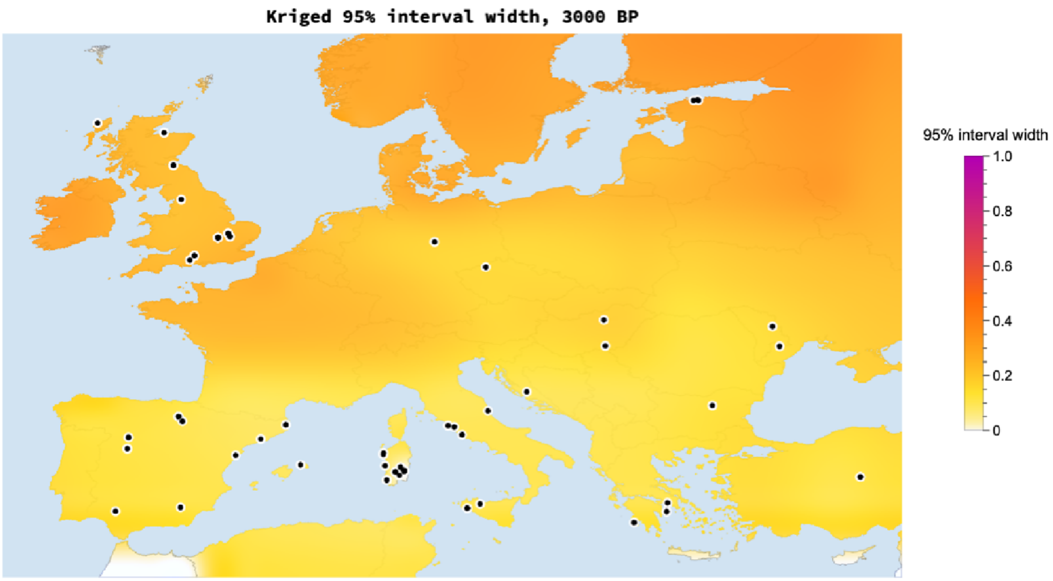
In[∘]:=
```
spatialOutputs = ExportSpatialVisualizations[Directory[], samples, grid, draws];
Import[spatialOutputs["MeanMap"]]
```

Out[∘]=



In[∘]:=
```
Import[spatialOutputs["UncertaintyMap"]]
```

Out[∘]=

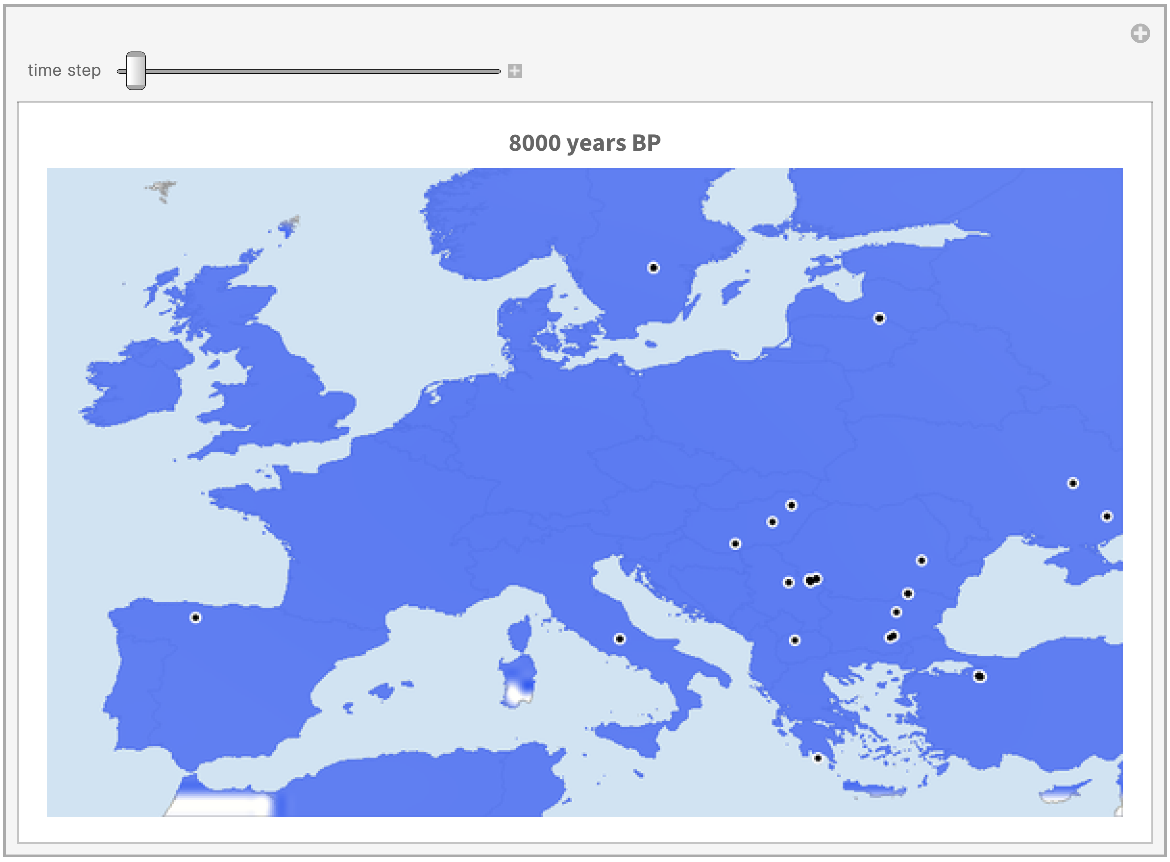


Posterior mean (top) and 95% credible-interval width (bottom) at 3000 BP. The two maps are a pair on purpose: the second is the error bar of the first, rendered at equal visual rank. The interval is widest exactly where the sampling map of §1 is emptiest — the model has the good manners to be unsure where it has seen nothing.

And because a static pair understates how the story unfolds, step through it yourself — each frame is the posterior mean with the ancient samples of that moment:

In[ ]:=
```
SpatialTimeExplorer[samples, grid, draws]
```

Out[ ]=



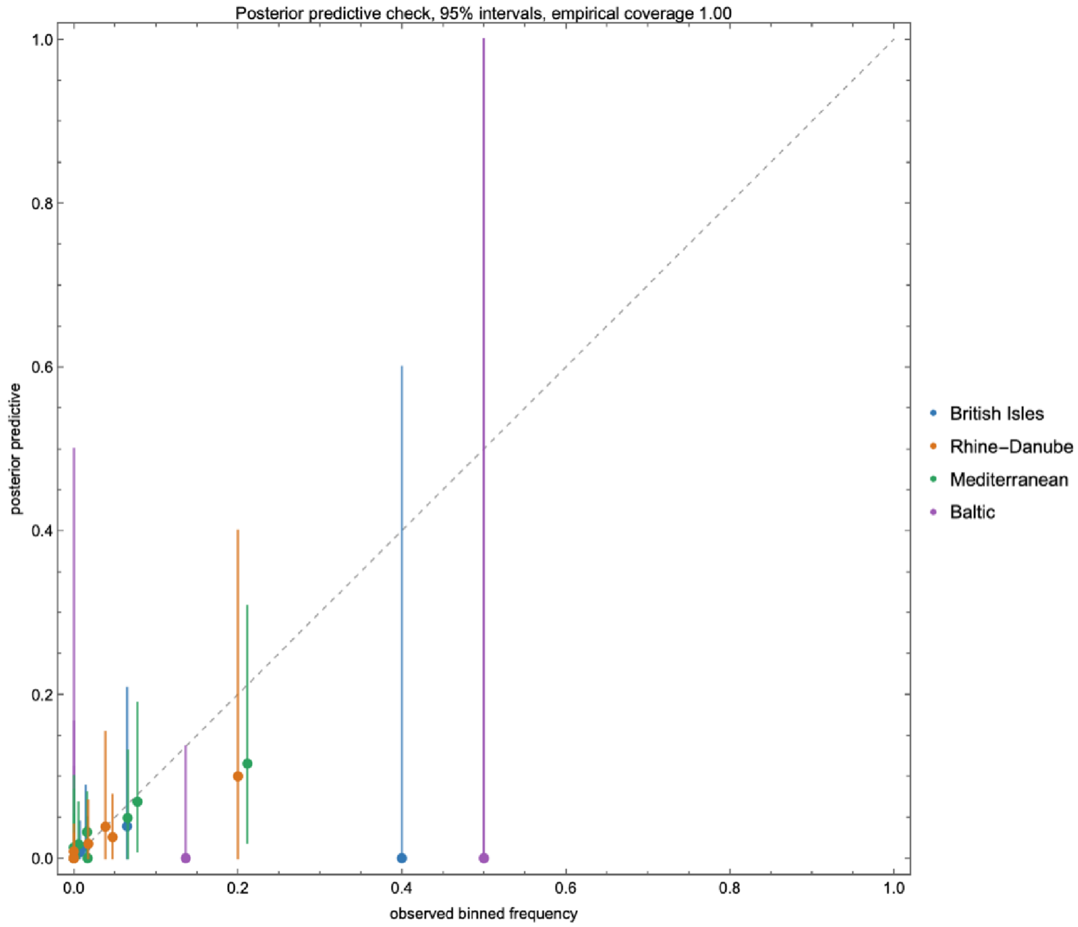
8. Does it actually predict anything?

A posterior that merely redescribes its training data is a compression algorithm, not a model. Three checks, in increasing order of severity.

Posterior predictive. Push posterior draws through the model, sample binomial allele counts at each bin's actual sample size, and ask how often the observed frequency lands inside the 95% band: 100% of the 37 bins. Bins with two called alleles correctly get bands covering nearly everything — wide honesty beats narrow fiction.

In[ ]:= **Import[smcOutputs["PosteriorPredictiveFigure"]]**

Out[ ]=



Held-out regions. Refit the entire SMC from scratch four times, each time deleting one region's samples, then predict the deleted region:

In[ ]:= **Dataset[Map[Association, Normal[Import[FileNameJoin[{Directory[], "data", "processed", "cross_valida**

Out[ ]=

HeldOutRegion	HeldOutBins	RMSE	Coverage95	FinalEpsilon	TotalSimula
British Isles	5	0.0896842	1	0.0501714	840
Rhine–Danube	10	0.0649381	1	0.0493997	840
Mediterranean	13	0.0236227	1	0.0466123	1320
Baltic	9	0.240045	1	0.0434562	1080

Held-out time — the killer test. Train only on samples older than 2,500 BP (1543 of them) and predict the 7 most recent bins. The intervals still cover, but the posterior median under-predicts the final rise badly (RMSE 0.32). Far from being embarrassing, this is the pipeline independently rediscovering Burger et al.'s headline: the evidence for *strong, late* selection is concentrated in the last three millennia. Delete those millennia and the model, sensibly, refuses to believe in them.

In[ ]:= Dataset[Map[Association, Normal[Import[FileNameJoin[{Directory[], "data", "processed", "time_slice_v

Out[ ]=

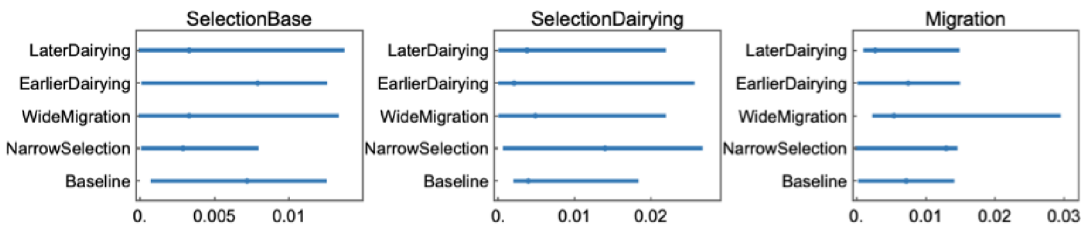
CutBP	HeldOutBins	TrainingSamples	RMSE	Coverage95	TotalSimulations
2500	7	1543	0.31855	1	1080

9. Which conclusions survive the priors?

Five complete refits: baseline; halved selection priors; doubled migration prior; and every dairying onset shifted 400 years earlier or later. The dairying-modulated selection component stays positive in all five (medians 0.008–0.019) — that conclusion belongs to the data. The migration posterior, by contrast, faithfully tracks whichever prior it is given (median 0.010 baseline → 0.020 under the widened prior): with 191 cells and regionally-pooled summaries, movement is weakly identified, and the honest statement is exactly that.

In[ ]:= Import[FileNameJoin[{Directory[], "figures", "generated", "sensitivity_intervals.png"}]]

Out[ ]=



10. Where did the allele come from? Origin versus selection

Everything so far answers "where did selection push hardest?" — which is a different question from "where did the allele come from?", and the two are easy to conflate when watching the animation. The classic origin estimate is **Itan et al. (2009)**, whose demic coevolution simulations placed the start of the lactase-persistence/dairying story around 7,500 years ago in the zone between the central Balkans and central Europe — roughly the Linearbandkeramik world of modern Hungary, Slovakia, and Austria. What do the ancient genotypes in this dataset say about first appearances?

In[ ]:=
```
earliest = TakeLargestBy[Select[samples, #["DerivedAlleles"] > 0 &], #["MeanDateBP"] &, 8];
Grid[
 Prepend[
  {Round[#["MeanDateBP"]], #["Country"], #["Latitude"], #["Longitude"],
    #["RS4988235Genotype"], #["Publication"]} & /@ earliest,
  Style[#, Bold] & /@ {"years BP", "country", "lat", "lon", "call", "publication"}],
 Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.2, 0.6},
 Background -> {None, {GrayLevel[0.93], None}}]
```

Out[ ]=

years BP	country	lat	lon	call	publication
6561	Bulgaria	43.2131	27.8644	GA	MathiesonNature2018
5600	Ukraine	49.5407	37.698	GA	MathiesonNature2018
5389	Romania	46.7877	23.5209	GA	GonzalesFortesCurrentBiology2017
5131	Ireland	54.5422	−5.95667	GA	CassidyPNAS2016
4466	Afghanistan	36.7833	70.	GA	NarasimhanPattersonScience2019
4350	Spain	42.4	−3.75	GT	OlaldeNature2018
4312	Great Britain	56.6809	−6.45982	GA	OlaldeNature2018
4270	Hungary	47.3832	19.0203	GA	OlaldeNature2018

In[ ]:=
```
GeoGraphics[
 Flatten[Table[{RGBColor[0.8, 0.15, 0.1], PointSize[0.013],
    Point[GeoPosition[{e["Latitude"], e["Longitude"]}]],
    Black, Text[Style[ToString[Round[e["MeanDateBP"]]] <> " BP", 9.5, Bold],
     GeoPosition[{e["Latitude"] + 1.1, e["Longitude"]}]]},
   {e, earliest}], 1],
 GeoRange -> {{34, 63}, {-14, 72}}, GeoProjection -> "Equirectangular",
 GeoBackground -> GeoStyling[{"CountryBorders",
   "Land" -> GrayLevel[0.97], "Ocean" -> RGBColor[0.85, 0.9, 0.95]}],
 ImageSize -> 640]
```

Out[ ]=

The eight oldest derived-allele carriers in the dataset. The earliest observations cluster in the southeast — Bulgaria ~6560 BP, Ukraine ~5600 BP, Romania ~5390 BP — before Ireland, Iberia, Britain, and Hungary appear around 5100–4270 BP. Each is a single heterozygote, so each is individually weak evidence, and sampling is far from spatially uniform — but the pattern is at least compatible with a central/southeastern European origin of the kind Itan et al. proposed.

The hero animation cannot show any of this, *by construction*: its model starts at 10,000 BP with a near-uniform initial frequency, has no mutation event, and — with migration weakly identified — moves essentially by in-place logistic growth. Britain lights up first there because its late samples demand the fastest growth, not because the allele originated there. So let us stop hand-waving and put the origin *into* the model: replace the uniform start with a point source — four new parameters (origin latitude, longitude, time, and the injection frequency), a wider migration prior (a travelling wave has to actually travel: m ∈ [0.02, 0.6] per generation, integrated stably with exact per-step logistic growth and exponential mixing), and diagonal adjacency so the wave can cross the Dover strait. The same SMC machinery then turns "where did it start?" into a posterior distribution:

In[∘]:=

```
originSmc = LoadOrRunOriginSMCABC[Directory[], samples, grid];
Grid[
 Prepend[
  Table[With[{q = SelectFirst[PosteriorParameterQuantiles[originSmc], #["Parameter"] === p &]},
    {p, NumberForm[q["Median"], {6, 3}],
     Row[{"[", NumberForm[q["Lower95"], {6, 3}], ", ", NumberForm[q["Upper95"], {6, 3}], "]"}]}],
   {p, {"OriginLatitude", "OriginLongitude", "OriginTimeBP",
     "Log10InjectFrequency", "Migration", "SelectionDairying"}}],
  Style[#, Bold] & /@ {"parameter", "median", "95% interval"}],
 Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.5, 0.6},
 Background -> {None, {GrayLevel[0.93], None}}]
```

Out[∘]=

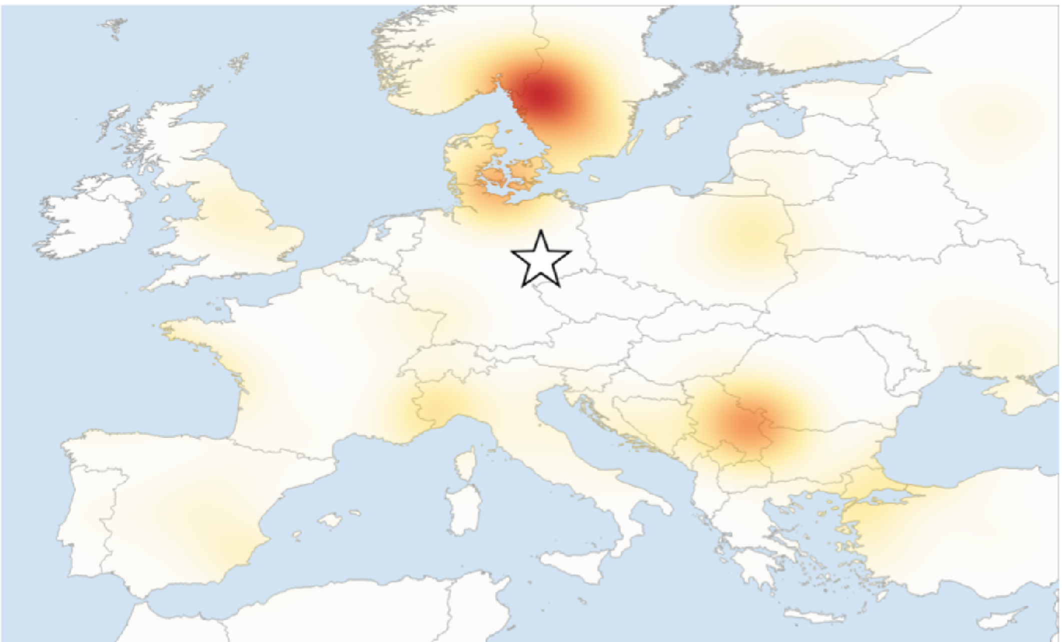
parameter	median	95% interval
OriginLatitude	51.488	[38.113, 61.773]
OriginLongitude	12.557	[−8.154, 32.956]
OriginTimeBP	8430.150	[6871.060, 9469.440]
Log10InjectFrequency	−2.153	[−2.966, −1.054]
Migration	0.306	[0.043, 0.591]
SelectionDairying	0.030	[0.016, 0.051]

In[∘]:=

```
OriginDensityMap[originSmc]
```

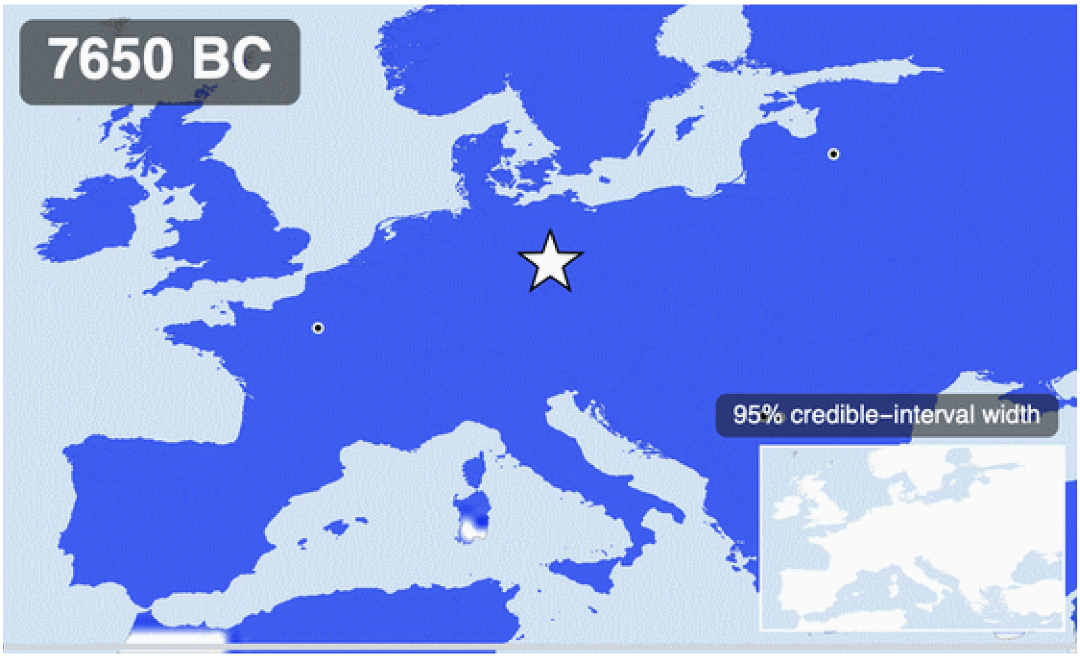
Out[∘]=

Posterior density of the allele's origin (star: weighted median)



The reverse-diffusion result, in the spirit of Itan et al.'s famous origin map: weighted posterior density of the point source's location, land-masked, with the weighted median starred. Where this mass sits — and how broadly it spreads — is the honest answer this dataset gives to "where did the rise begin?".

And the forward story from that fitted origin — the travelling wave the hero animation could not show — as its own video (the star marks the posterior-median origin; the inset again tracks the 95% credible-interval width):



Forward simulation from the fitted point source: posterior mean over 100 origin–model draws, 100–year interpolated steps from 9,600 BP. Generated by ExportOriginSpread; the H.264 version is figures/generated/origin_spread.mp4.

Read the origin posterior with the same discipline as everything else here. The location information comes from a handful of early single-heterozygote calls scattered over an unevenly sampled map, filtered through a coarse deterministic wave model — so the credible region is wide, and it should be. What the fit adds beyond the anecdotal table above is that all three ingredients are now estimated *jointly*: an origin location and time, a wave that must actually reach Britain and the Baltic in time for their observed rises, and selection on top. Whether the posterior mass sits over the Carpathian Basin, the lower Danube, or the Pontic steppe is exactly the kind of statement this machinery exists to make — and unlike a caption, it comes with its uncertainty attached.

11. The animation, assembled

The hero animation at the top of the post is the whole pipeline in one artifact: 81 time-interpolated frames of the kriged posterior mean, the moving window of ancient samples, a progress bar, and — non-negotiably — the credible-interval inset, so that even a viewer who reads nothing else cannot mistake the smooth colours for certainty. Because every frame is raster-composited over a single cached base map, the full render takes under a minute:

```
hero = ExportHeroAnimation[Directory[], samples, grid, draws];
(* figures/generated/hero_lactase_persistence.{mp4, gif} *)
```

12. What this is not

This is a calibrated caricature, and the caveats are part of the result. The regional logistic layer reproduces the published four-region picture qualitatively, not parameter-for-parameter. The simulator is deterministic — no genetic drift — which flatters the posterior's tightness; demography (the steppe migrations!) is implicit; Ireland is unreachable in the adjacency; radiocarbon uncertainty enters the bins as point dates; the final ESS of 26.6 keeps every interval approximate; and everything after ~2,000 BP extrapolates beyond the data. Next steps, in rough order of value: a formal model comparison between the uniform-start and point-source variants (§10), a Wright–Fisher stochastic core, sample-level ancestry-aware likelihoods, MCMC rejuvenation inside the SMC, and calibrated-date uncertainty propagated end-to-end.

13. Reproducibility

Everything — data retrieval with checksums, processing with provenance, fits, SMC, validation, sensitivity, every figure, the hero animation, and this notebook itself — regenerates from the repository:

```
git clone https://github.com/mthiel74/ancient-dna-lactase-spatial-wolfram
wolframscript -file scripts/run_pipeline.wls --particles 400 --generations 5
wolframscript -file scripts/run_tests.wls
wolframscript -file community/build_notebook.wls
```

The package `src/LactasePersistenceSpatial.wl` (~2,000 lines) carries the entire workflow behind the one-liners in this post; 24 VerificationTests cover the parsers, the simulator, the summary statistics, and the SMC machinery, and GitHub Actions runs them non-interactively. Repository: mthiel74/ancient-dna-lactase-spatial-wolfram.

14. References

Evershed, R. P., et al. (2022). Dairying, diseases and the evolution of lactase persistence in Europe. *Nature* 608, 336–345. nature.com/articles/s41586-022-05010-7

Burger, J., et al. (2020). Low prevalence of lactase persistence in Bronze Age Europe indicates ongoing strong selection over the last 3,000 years. *Current Biology* 30(21), 4307–4315. cell.com/current-biology

Itan, Y., Powell, A., Beaumont, M. A., Burger, J., Thomas, M. G. (2009). The origins of lactase persistence in Europe. *PLoS Computational Biology* 5(8), e1000491.

GLAD LP Ancient Genotypes 2022 workbook (UCL), derived from the Allen Ancient DNA Resource v44.3.

Analysis, figures, and notebook generated with Wolfram Language 15.0.1 on August 31, 2026.